

DECT-2020 NR for Energy and Power Grid Substation QoS Monitoring Use Cases

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Abstract—Modern power grids and industrial systems require scalable and reliable wireless connectivity to support real-time monitoring and control. DECT-2020 New Radio (NR), an ETSI-standardized wireless technology operating in license-exempt spectrum, offers native mesh networking, low latency, and flexible deployment for industrial Internet-of-Things (IoT) applications. This paper investigates the suitability of DECT-2020 NR mesh networking for power grid monitoring within the Ultra Scalable Wireless Access (USWA) project. Two proof-of-concept (PoC) deployments are evaluated, collecting data from multiple sensors under varying conditions, including packet fragmentation, multi-hop communication, and different data rates to assess the end-to-end performance of the system in terms of quality-of-service (QoS) metrics. From the results, we found that DECT-2020 NR shows potential for larger scale energy and substation monitoring deployments.

Keywords—DECT-2020 NR, Internet of Things (IoT), Proof of Concept (PoC), Mesh Networks

I. INTRODUCTION

Digitization of industries requires efficient, flexible, and cost-effective wireless connectivity solutions. Many industrial facilities still rely on cables for communication, which limits scalability and reconfiguration. DECT-2020 New Radio (NR) – a modern wireless ETSI standard – has emerged as a promising candidate to address this gap. DECT-2020 NR offers a state-of-the-art radio interface tailored for industrial Internet-of-Things (IoT) applications, including native mesh networking capabilities that enable large-scale local networks through multi-hop relaying between devices, as well as point-to-point and star network topologies. Operating in a license-exempt band (the frequency band previously used by Digital Enhanced Cordless Telecommunication devices, i.e., legacy DECT), it allows industrial users to deploy and manage private networks without licensing costs, opening opportunities for more autonomous and scalable digitization efforts in factories, buildings, utility sites, and cities.

This paper is conducted within the framework of the Ultra Scalable Wireless Access (USWA) project (project running from 12/2022 to 11/2025), which investigated how advanced DECT-2020 NR features can support diverse industrial use cases. The project goals included defining system architectures and performance requirements, developing scalable and energy-efficient mesh networking for massive machine type communications (mMTC) IoT deployments, enabling ultra-reliable low-latency communications (URLLC) for time-critical applications, and validating the technology through real-world proof-of-concept

demonstrations. Together, these efforts aim to advance DECT-2020 NR as a robust and flexible solution for next-generation industrial wireless networks.

Wireless private networks are well-suited for modern energy-grid communications because they provide the near-real-time, reliable, and flexible connectivity that renewable and distributed energy systems require. As power grids integrate more volatile sources, such as wind and solar, wireless communication enables fast monitoring and control, helping to maintain grid stability. It also supports smart-grid functions, such as smart metering, automated control, and fault detection, by offering low latency and wide-area coverage. Compared to wired communications, wireless networks are easier and cheaper to deploy in diverse grid environments, making them more scalable as the grid expands [1]. Furthermore, as the power grid is often targeted in military conflicts, wireless networks can support observability if they are decoupled from the wired infrastructure.

The main contributions of this paper are threefold. First, it explores the applicability of DECT-2020 NR mesh networking to energy grid monitoring, with a focus on meeting the stringent requirements of modern smart grid communications. Second, it presents a detailed system design and implementation of a proof-of-concept testbed for Electrical Network QoS Monitoring (ENQM), integrating DECT-2020 NR with 5G backhaul and a suite of environmental and electrical sensors, based on the system presented in [2]. Third, it provides experimental results that evaluate the performance of the proposed system in terms of delay, jitter, and data rate across multiple use cases. These contributions study the potential of DECT-2020 NR for scalability, reliability, and energy-efficiency in monitoring of future energy grids. To our knowledge no other studies have been conducted in this topic.

The remainder of this paper is structured as follows. Section II reviews recent research and technical background relevant to DECT-2020 NR and its applicability to energy monitoring and industrial IoT scenarios. Section III details the mesh operation principles of DECT-2020 NR, including routing and forwarding mechanisms. Section IV presents specific use cases, considerations for system design, and proof-of-concept implementations within the context of energy quality monitoring. Section V discusses the experimental results obtained from these implementations and highlights key performance metrics. Finally, Section VI concludes the paper by summarizing the findings and presenting opportunities for future work.

II. BACKGROUND

Recent research on DECT-2020 NR as a candidate technology for massive machine-type communications (mMTC) has mainly focused on system-level and link-level simulations to assess its scalability, reliability, and compliance with ITU requirements. Anttonen et al. [3] conducted comprehensive NS-3 simulations to evaluate the performance of DECT-2020 NR mesh networks in urban macro environments. Their results demonstrate that the technology can support device densities well above one million devices per square kilometre, while maintaining packet outage rates below 1% and end-to-end delays under 10 seconds. The study highlights the flexibility of mesh routing, where radio devices autonomously relay traffic, and clusters may operate on different channels to mitigate interference. The authors also note that DECT-2020 NR is able to fulfil the ITU requirements for mMTC, making it a strong candidate for IMT-2020-enabled digital services.

Kovalchukov et al. [4] further support these findings through system-level simulations, comparing DECT-2020 NR with traditional LPWAN solutions. Their work shows that DECT-2020 NR outperforms other technologies in terms of supported node density and delay/loss guaranties, thanks to its decentralized mesh topology and dynamic channel selection. The authors emphasize the benefits of autonomous operation and multi-hop routing, which allow for flexible deployments without the need for centralized infrastructure or network planning.

Samuylov et al. [5] focus on the impact of MAC and physical layer mechanisms, such as energy conservation, relay selection, and power control, using system-level simulation tools. Their results indicate that network density and energy conservation parameters have the most significant effect on packet delivery performance, while relay selection and power control play a secondary role in dense networks. The study also finds that the listen-before-talk protocol and dynamic channel selection contribute to robust interference management, ensuring reliable operation even in challenging environments.

In the realm of power grid communications, IEC 61850 is

a widely adopted international standard that defines node-based object-oriented data models. Through the Abstract Communication Service Interface (ACSI), the data models are mapped to relevant protocols such as Manufacturing Message Specification (MMS), Generic Object-Oriented Substation Event (GOOSE), and Sampled Values (SV). These protocols serve different functions, while MMS supports client-server operations over TCP/IP and Ethernet, GOOSE and SV implement real-time messaging for protection, interlocking, and transmitting digitized current and voltage samples [6], [7]. The deterministic behavior for time-critical IEC 61850 messages is ensured through switched Ethernet with VLAN and priority tagging (IEEE 802.1Q/802.1p). According to Kokkonen-Tarkkanen et al. [8], fault location applications use the 9-2LE SV protocol in a point-to-point setup with a 250 ns signal interval, 3 ms latency, ~5 ms jitter, whereas edge-to-edge intertrip protection relies on R-GOOSE in a broadcast configuration with 1 s to 1 min signal intervals, 3 ms latency, and 10 ms jitter, both scenarios requiring ultra-high availability. MMS typically is reserved for log reports sent in fixed time intervals or event driven notifications and alarms. In [9] Van Rensburg et al. discuss IEC 61850 transfer time classes where delay requirements are either greater than 1000 ms if the messages are periodical updates (TT0) and from 500 ms to 1000 ms if they are event driven (TT1 and TT2). The data link layer MTU is not defined in the standards for MMS communications. Due to the stringent requirements for SV and GOOSE, the current commercial implementations of DECT-2020 NR are not able to meet the specified performance targets. Therefore, we study in this paper whether this technology is suitable for MMS type communications in monitoring use cases.

III. DECT-2020 MESH OPERATION ROUTING AND FORWARDING

The DECT-2020 New Radio standard introduces a flexible architecture for local wireless access networks, supporting both ultra-reliable low-latency communication (URLLC) and massive machine-type communication (mMTC) use cases. Its mesh topology is a key feature, due to its self-healing capabilities and its decentralized topology. As shown in Figure 1 mesh operation is based on a clustered tree structure, where Radio Devices (RDs) dynamically select upstream nodes using beacon information and routing cost metrics. Devices can operate in Fixed Termination (FT) mode, with backend connectivity acting as Sinks, or in Portable Termination (PT) mode associating to FT mode devices, or in both FT and PT mode simultaneously, forming multi-hop paths to extend coverage. Cluster formation occurs hop-by-hop, with each RD, in PT mode, associating to the most suitable parent RD, operating as FT, based on link quality and cost. This decentralized approach supports high device densities and robust connectivity in interference-prone environments while extending the network coverage [10].

Data routing and forwarding in DECT-2020 networks are implemented at the Data Link Control (DLC) layer, which provides packet routing services for uplink, downlink, and RD-to-RD communication. Uplink routing towards the backend uses hop-by-hop forwarding based on cluster tree topology formed and routing cost values, as discussed above, while downlink routing employs Selective Flooding, Selective

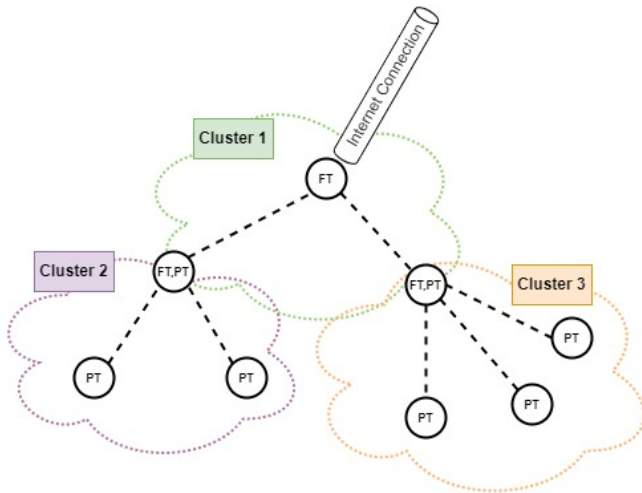


Fig. 1: Cluster tree topology

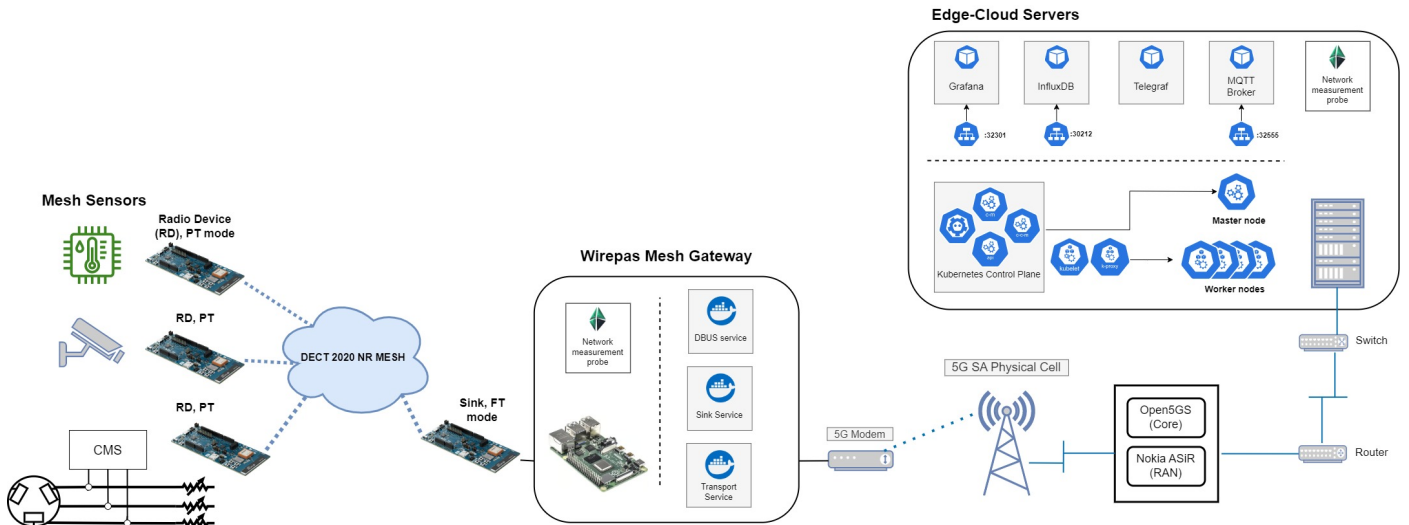


Fig. 2: ENQM PoC 1 implementation

Source Routing, or Cached Downlink Routing to optimize delivery in cluster tree. RD-to-RD communication relies on hop-limited flooding to minimize latency locally. The DLC layer supports multiple service types, including transparent mode, segmentation, and ARQ-based reliable transmission. Above DLC, the Convergence (CVG) layer offers end-to-end services such as segmentation/reassembly, flow control, lifetime management, and optional security. Multiplexing at CVG uses Endpoint (EP) identifiers or Mux Tags to distinguish application flows, enabling adaptation of higher-layer protocols such as IPv6 [11].

In DECT-2020, routing recovery from broken links or RDs going offline is handled through autonomous, local mechanisms such as continuous link-quality monitoring, fast local rerouting, and dynamic re-association to alternative neighbors. When a link or relay RD fails, affected nodes can independently select new next hops based on updated routing cost metrics, without requiring centralized coordination or global routing tables, while downlink delivery remains robust through selective flooding or source-assisted routing. This self-healing, QoS-aware mechanism minimizes disruption and signaling overhead, which is a key reason why DECT-2020 NR is considered a highly reliable solution for large-scale deployments with dense device populations and dynamic network conditions [10].

IV. ENQM USE CASES AND SYSTEM PoCs

In the context of energy grids, DECT-2020 has been identified as a technology with the potential to facilitate modular and low-maintenance deployment of monitoring components such as Intelligent Electronic Devices (IEDs) and environmental sensors. This paper builds upon the use cases introduced in [2] and presents two PoCs that aim to validate ENQM requirements namely latency and jitter values. As mentioned in Section II, DECT-2020 technology shows potential to meet the expected network performance indicators (KPIs) for electricity monitoring, while leveraging the benefits of mesh routing capabilities presented in Section III. Deployment and

maintenance costs are significantly reduced due to the absence of wires for its implementation and the low power consumption during operation (in the order of a hundred mA). Furthermore, as DECT-2020 NR operates in an unlicensed band, therefore deployments can be done without license or spectrum fees making it relatively inexpensive to deploy at scale. In light of DECT-2020's capabilities, there are strong prospects for this technology to advance from the PoC's presented here toward production deployment, in real power grid substations.

Two PoCs have been developed in this work. The first one, for simplicity named PoC 1, was developed in VTT's facilities in an indoor environment, and the second, named PoC 2, was developed in the Tampere University Campus in Hervanta. PoC 1 illustrated in Figure 2, presents a testbed that comprises three use cases (UC). The aim of these use cases is to collect, using various sensors, monitoring data from a power grid substation. UC 1 obtains environmental information using a temperature and humidity sensor (Vaisala H110 in the PoC), while UC 2 determines the person intrusion probability from a video sensor (applying YoLov4 CNN algorithm to a video stream). UC 3 consists on reading electricity consumption data via a three phase Circuit Monitoring System (CMS), connected to a server room's power distribution system in the testbed. All sensor readings are then forwarded to a display at a remote monitoring center, that would typically be operated by a Transmission System Operator (TSO) or Distribution System Operator (DSO).

The PoC 1 implementation integrates multiple technology platforms. From left to right of Figure 2, the sensors connect to a Wirepas 5G mesh with six Nordic Semiconductor nrf9160-dk chipsets and a gateway that implements DECT NR using band 9 (1 910 to 1 930 MHz for both reception and transmission). The gateway then forwards the monitoring data to an edge cloud backend through a 5G SA network. The backend, through cloud technologies (Telegraf, InfluxDB and Grafana), displays live metrics in a dashboard. In this manner, PoC 1 is able to emulate these use cases in an indoor laboratory setup.

For evaluating the ENQM network requirements, two tools have been used. Qosium network probes have been installed in

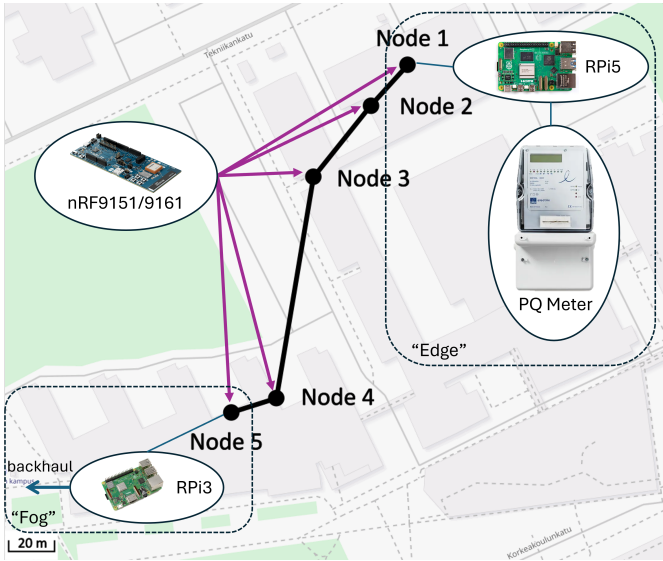


Fig. 3: PoC 2: Power Quality measurements over DECT-2020 NR mesh network at the Tampere University Campus.

the mesh gateway and in the edge cloud servers for measuring IP-based traffic through the cellular 5G and Edge Cloud networks. For traffic routed through the mesh network the Wirepas Gateway APIs provided relevant metrics such as uplink latency, QoS and hop count.

In PoC 2, a set of five DECT-2020 NR capable nodes (nRF9151/nRF9161) were configured as a 4-hop mesh network as shown in Fig. 3. In this scenario, a Power Quality (PQ) meter was used to monitor and record voltage dip/swell and rapid voltage change events at the "Edge". These events were then polled once a minute and sent through the mesh network as JSON file. One Raspberry Pi 5 (RPi5) device was used to convert and push the data from the PQ meter to the nodes. The last node of the mesh network, the sink node, was used to connect the mesh network devices into a gateway (Raspberry Pi 3, RPi3) that then made a connection to backhaul (internet/server) possible.

The second PoC sends event updates using a lightweight publish-subscribe messaging protocol Message Queuing Telemetry Transport (MQTT) in order to communicate to a server. In order to get the messages through the mesh network, the JSON files were segmented at DECT-2020 NR CVG layer in order to fit them to the 180 byte limit defined by implementation for the mesh network packets. In practice, the JSON file sizes from the PQ meter were in the order of 1170 bytes, meaning multiple packets had to be sent, resulting in increased total latency. The latency was also studied and compared by having only a 1-hop device-to-device scenario to the larger 4-hops mesh configuration. This consisted of sending one PQ meter JSON file per minute over a 48-hour time period.

TABLE I: Mean mesh network KPI values per UC, 2 hops

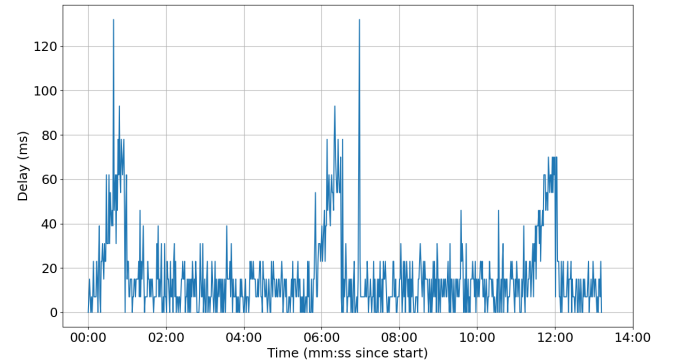
Mean Values	UC 1	UC 2	UC 3
Latency (ms)	15.45	18.55	27.93
Jitter (ms)	11.0	20.55	14.84
Datarate (kbps)	0.736	0.468	14.581

TABLE II: End-to-end network KPI values per UC

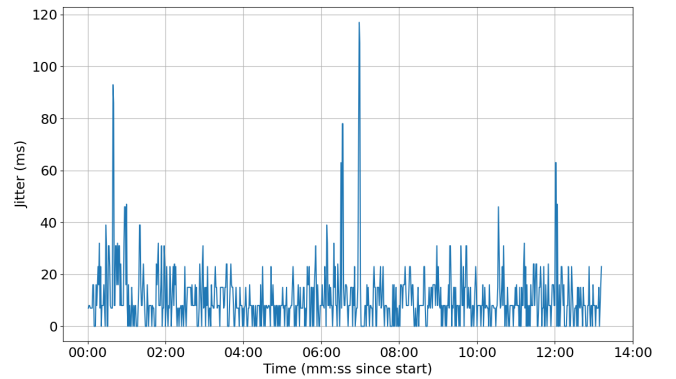
Mean Values	UC 1	UC 2	UC 3
Latency (ms)	26.6	29.71	39.62
Jitter (ms)	13.95	23.52	17.83
Datarate (kbps)	3.95	3.719	50.577

V. RESULTS

This section presents the results obtained from the two PoCs introduced in Section IV.



(a) 3 hop Delay



(b) 3 hop Jitter

Fig. 4: Timeseries mesh network measurement results for 3 hops in PoC 1

A. PoC 1 Results

For the first PoC three use cases (UC) were presented for the purposes of power substation monitoring UC 1 for temperature and humidity, UC 2 for video intrusion detection, and UC 3 for electricity QoS monitoring. Two separate measurements had to be performed for each use case to obtain the end-to-end performance of the use case due to the differences in forwarding and transport layer technologies between the DECT 2020 NR mesh and the 5G SA network. Two result tables are presented for comparing the use cases under the same conditions (2 hops): Table I shows mesh network key performance metrics for each UC, and Table II shows the end-to-end use case values covering both mesh and 5G cellular networks.

In PoC 1 each UC has different traffic profiles in terms of sampling and data rates. UC 1 and UC 2 have a sampling rate of 1 message per second with payloads of 92 and 66 bytes respectively (excluding the DECT-2020 NR protocol headers), while UC 3 has an average payload size of 175 bytes and a sampling rate of 10 Hz, meaning some messages can exceed the maximum DECT NR MTU size (180 bytes) and therefore can be fragmented at CVG layer of DECT-2020 NR protocol stack. When evaluating the end-to-end metrics messages are sent over MQTT to the edge servers. This process adds data to the payload and new headers therefore increasing the overall data rate of the application as shown in Table II. The mean values for the mesh latency and jitter from Table I are added to the mean values obtained from the cellular 5G RAN and backhaul communications resulting in the latency and jitter values obtained in Table II.

Additionally in the first PoC, mesh network performance over multiple hops was evaluated. Figures 4a and 4b show a time series measurement of UC 1 traffic (sampling rate 1Hz, 92 bytes payload) over three hops during 13 minutes. Mean latency values for one, two and three hops are 14.36, 15.45, and 18.739 ms respectively. While mean jitter values are 9.8, 11.0 and 14.854 ms for one, two and three hops respectively. The standard deviation of UC 1 traffic over 3 hops reaches 48.54 ms, indicating that messages can arrive at the gateway with a delay well over the mean. Finally, during the numerous measurement carried out for the PoC 1 use cases, no packets were lost showing 100% service availability.

B. PoC 2 Results

The latency of PoC 2 was measured with synchronized end devices (RPi3/RPi5) and 1-hop versus 4-hops results were collected over the whole chain. The results presented in Fig. 5 and Fig. 6 show how the total application layer end-to-end latency behaves. The statistics of the latency and jitter values are presented in Table III. It should be noted that one packet of the total of 2880 packets reached a maximum latency of 379.5 ms, however, this outlier is omitted from Fig. 5 to improve graphical readability. This is also visible in Fig. 6 as the only high jitter peak. No packets were lost resulting to a success rate of 100 %.

TABLE III: PoC 2: 1-hop and 4-hop end-to-end latency and jitter results (combined latency of a large JSON file splitted into several packets).

Configuration	Maximum	Minimum	Mean	STD
Latency: 1-hop (ms)	207.28	127.50	153.71	11.02
Jitter: 1-hop (ms)	49.62	0.0020	9.39	7.15
Latency: 4-hop (ms)	379.54	149.80	171.40	13.06
Jitter: 4-hops (ms)	202.55	0.0010	12.48	11.12

VI. CONCLUSIONS

The two PoCs developed in this work highlight how the different use cases presented may have varying performance when using the DECT 2020 mesh network. The major factor for the latency in message delivery, is the fragmentation of JSON packets. While in PoC 1 packets are not fragmented and delays remain under 50 ms, in PoC 2 fragmenting packets in seven segments results in delays over 100 ms. The impact of adding

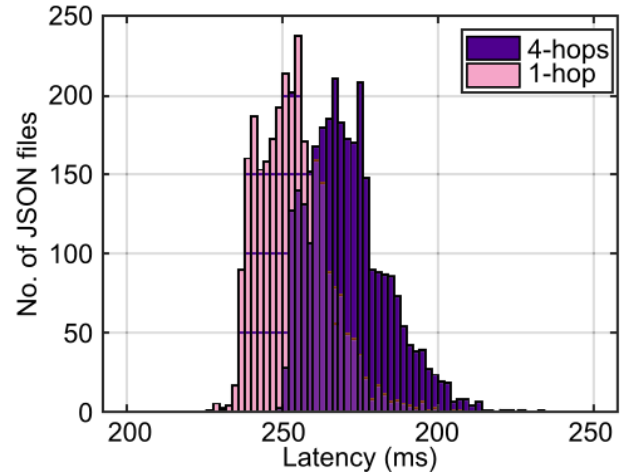


Fig. 5: PoC 2: Histogram of the 1-hop and 4-hop end-to-end latency results (total time to deliver JSON files with multiple packets over the mesh network).

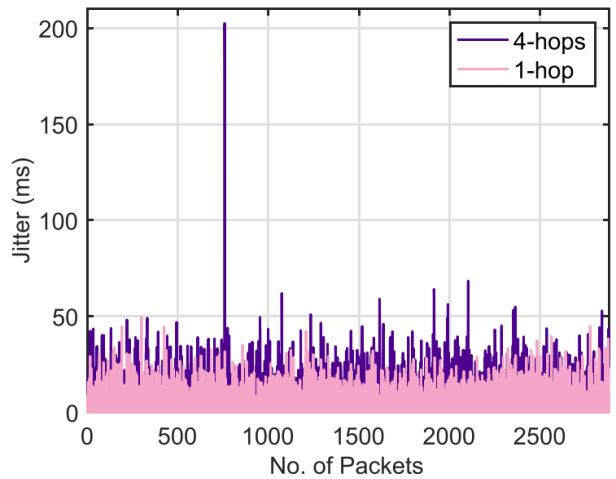


Fig. 6: PoC 2: 1-hop and 4-hop end-to-end jitter results.

more hops is noticeable but minor compared to fragmentation. Additionally, an aspect that makes this technology apt for monitoring is the reliability in terms of packet loss. Both PoC 1 and PoC 2 have shown no packet drops, even in experiments lasting two days.

According to the IEC 61850 transfer time requirements outlined in [9], for periodic message updates and non-protection event driven messages delay values ranging from 500 ms to 1000 ms are sufficient. Nevertheless, the results show that it's important to determine which values are relevant for monitoring purposes. For instance, in PoC 1 the CMS-700 device, used for monitoring energy QoS, may use 96 sensors which have several dozens of parameters [12]. Trying to monitor all the possible parameters from a single IED will be of no use since link delays will increase and may cause network congestion. Integration engineers may consider decreasing the RD to gateway ratio to mitigate congestion delays in the mesh LAN and using

additional links with higher capacity for the backend (for example, cellular 5G as shown in Figure 2).

Additionally the PoCs have demonstrated the end-to-end system by incorporating a database and a graphical user interface (GUI) using InfluxDB and Grafana. Updates to the Grafana monitoring dashboard come every one second, therefore the DECT-2020 NR mesh network can deliver data well within that time interval. DECT-2020 NR shows potential for larger scale energy and substation monitoring deployments. In future work, we plan to build upon these results by extending the evaluation to wider deployments and more complex grid environments for long-term operational performance.

ACKNOWLEDGMENT

This work has been performed in the framework of European Celtic-Next project Ultra Scalable Wireless Access (USWA). Finnish partners acknowledge funding by the European Union - NextGenerationEU - received through Business Finland.

REFERENCES

- [1] Huawei Enterprise Wireless, "Power grid wireless private network," <https://e.huawei.com/en/blogs/enterprise-wireless/2024/power-grid-wireless-private-network>, 2024, accessed: 2025-11-25.
- [2] J. S  e, R. Pires, M. Lasanen, J. Pirskanen, M. Alam, H. Ahmad, A. Bin, E. Llaguno, I. Pretel, and M. Valkama, "State of play and results of the uswa project," 06 2025.
- [3] A. Anttonen, P. Karhula, M. Lasanen, and M. Majanen, *Enabling Massive Machine Type Communications with DECT-2020 Standard: A System-Level Performance Study*, ser. VTT Research Report. Finland: VTT Technical Research Centre of Finland, 2021, no. VTT-R-00367-21, vTT-R-00367-21 (lis  etty tiedosto Antti Anttosen pyynn  st  i),128181/SySimD2020.
- [4] ETSI, "DECT-2020 New Radio (NR); Part 1: Overview; Release 1," *European Telecommunications Standards Institute, Technical Specification (TS)*, vol. 103, pp. 636–1, March, 2024.
- [5] A. Samuylov, D. Moltchanov, M. Alam, J. Pirskanen, J. Numminen, M. Valkama, Y. Koucheryavy, and M. Valkama, "Performance of mac layer mechanisms in dect-2020 nr mmhc technology," in *2024 IEEE 99th Vehicular Technology Conference (VTC2024-Spring)*, 2024, pp. 1–6.
- [6] C. Brunner, "Iec 61850 for power system communication," *IEEE Transactions on Power Delivery*, vol. 23, no. 1, pp. 618–624, 2008.
- [7] A. Inc., "New standard lets data center devices change functions and talk to all their neighbors," ABB, Tech. Rep., 2015, white Paper. [Online]. Available: <https://www.abb.com/datacenters>
- [8] H. Kokkonen-Tarkkanen, P. Raussi, S. Horsmanheimo, P. Hovila, A. Kulmala, and S. Borenius, "5g edge for power system applications," in *27th International Conference on Electricity Distribution (CIRED 2023)*, vol. 2023, 2023, pp. 1871–1875.
- [9] M. van Rensburg, D. Dolezilek, and J. Dearien, "Case study: Using IEC 61850 network engineering guideline test procedures to diagnose and analyze ethernet network installations," in *Proceedings of the PAC World Africa Conference*, Johannesburg, South Africa, Nov. 2015. [Online]. Available: <https://selinc.com/api/download/112218/>
- [10] *DECT-2020 New Radio (NR); Part 1: Overview; Release 2*, ETSI Std. TS 103 636-1 V2.1.1, October 2024, available: <https://www.etsi.org/standards-search>.
- [11] *DECT-2020 New Radio (NR); Part 5: DLC and Convergence layers; Release 2*, ETSI Std. TS 103 636-5 V2.1.1, October 2024, available: <https://www.etsi.org/standards-search>.
- [12] ABB, *CMS Manual*, Online, ABB, Zurich, Switzerland, 2024, manual for the ABB CMS-700 device. [Online]. Available: https://library.e.abb.com/public/ae914775dba84d5ea4ed896c6c0378f9/CMS%20Manual_web.pdf